

Shelby Pullen

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EDUCATION

- **University of California, San Diego** San Diego, CA
PhD student - Aerospace Engineering *September 2025 - May 2030*
- **George Washington University** Washington, DC
Bachelor of Science - Mechanical Engineering - Summa Cum Laude *August 2021- May 2025*
Concentration in Aerospace Engineering, Minor in Mathematics
Pi Tau Sigma Honors Society - Phi Gamma Chapter

SKILLS SUMMARY

- **Languages:** MATLAB (Parallel Computing toolbox, Optimization toolbox), Python
- **Softwares:** Fusion 360, CAD and CAM), Solid Works CAD and FEA, AutoCAD, Microsoft Excel, EnergyPlus
- **Soft Skills:** Leadership, Event Management, Writing, Public Speaking, Time Management

RESEARCH EXPERIENCE

- **UC San Diego: Dr. Boris Kramer's Lab** San Diego, CA
Graduate Student Researcher *September 2025 - Present*
 - Developed a gradient based optimization algorithm for designing meta-materials for high strain rate impact conditions
 - Extended deterministic framework to incorporate parameter uncertainty via Monte Carlo sampling across for robust design optimization of meta-materials represented by a chain of masses, dampers, and nonlinear springs
 - Implemented gradient-based global optimization using MATLAB's MultiStart and fmincon with nonlinear constraints across a bounded parameter space
 - Configured MATLAB Parallel Computing Toolbox for multi-core execution on HPC systems
- **GWU: Dr. LeBlanc's Material Science and Additive Manufacturing Lab** Washington, DC
Undergraduate Research Assistant *May 2024 - May 2025*
 - Created micro scale surface textured samples inspired by biological systems including seal fur and owl down to test their reduction in hydrodynamic noise along the hull of submarines to minimize detection
 - Utilized Stereolithography 3D printing (SLA) to manufacture samples inspired by seal fur for testing in a water channel for high resolution
 - Conducted literature reviews focused on boundary layer vibration production, biological surface texture effects on pressure drag, and micro-texture production using imprinting and additive manufacturing
- **GWU: Dr. LeBlanc's Energy Engineering Research Lab** Washington, DC
Undergraduate Research Assistant *June 2023 - May 2024*
 - Independently conducted a research project for Homeland Security to find electrical load estimates for each zip code in DC leveraging open data sources and EnergyPlus
 - Worked on Naval De-carbonization project to lower carbon emissions on Naval ships using energy device configurations using alternate fuel efficiencies for naval ships
 - Implimented the Python libraries, NumPy and Pandas, to clean property tax data to produce an accurate count of each building type in each zip code
 - Simulated the energy load for each building type for every hour in a year in EnergyPlus, utilizing local weather data

WORK EXPERIENCE

- **Naval Surface Warfare Center Carderock Division** Bethesda, MD
Intern - Advanced Manufacturing Lab - Secret Clearance *May 2025 - August 2025*
 - Built a toolkit for welding engineers in Python and a graphical user interface to determine weld parameters and conditions:
 - * Required preheat settings concurrent with Navy Technical Specifications for submarine manufacturing
 - * Cooling rate isotherm graphs and CCT diagrams with the cooling curve overlayed to predict material structure
 - * Dilution percent and resulting chemical composition for additive manufacturing (AM) and traditional welding
 - Derived a new dilution equation for hybrid twin wire welding to predict weld bead geometry and chemical composition
 - Simulated chemical compositions for future AM experimentation using proprietary alloys in hybrid twin wire welding
 - Processed and analyzed sensor data of Laser Hot Wire experiments of weld parameters for every bead, layer, and part

- **GW Baja SAE**

Co-Captain

Chief Engineer - 4WD and Manufacturing Team Lead

Team Member

Washington, DC

August 2024 - June 2025

August 2023 - August 2024

August 2022 - August 2023

- Designed and built an off roading vehicle around a new 14 hp engine and competed against other collegiate teams in international competitions
- Utilized SolidWorks to design and determine strength of new design for suspension and drivetrain parts using FEA
- Designed front suspension and steering using Optimum Kinematics and SolidWorks to optimize ride height, turning radius, and bump steer
- Led CNC manufacturing by operating CAM in Fusion to program a Haas fourth axis mini mill to create custom parts for the car: Hubs, spindles, and gearbox casing
- Remodeled, enhanced and implemented the four-wheel drive system using a chain drive and custom clutch with new attachment designs and proper tensioning

- **GWU: Department of Mechanical and Aerospace Engineering**

Washington, DC

Undergraduate Learning Assistant August 2023 - December 2023

- Personally helped students develop a better understanding of how to use Python for technical use for data and model analysis in and out of the classroom
- Created an inclusive environment for students to ask questions and demonstrate their knowledge on Data Frames, Python Functions, and Linear Algebra

PROJECTS

- **Senior Capstone Project - Drone Design for Prolonged Flight Time with Surveillance Applications:** Developed design for drone flight time of greater than 10 hours using solar panels and helium balloons for added lift (August 2024 - May 2025)

FELLOWSHIPS

- Greg Warner Fellowship - November 2025